HARSH JHA

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SUMMARY

Full-stack developer and AI engineer at **Voxomos**, building agentic AI, ASR evaluation, and **FastAPI/MySQL/MongoDB** systems — plus independent projects in computer vision, financial forecasting, and geospatial apps using **React** and **Python**.

EDUCATION

JSS Academy of Technical Education — **Oct 2022 – Jul 2026**
Bachelor of Technology in Information Technology — ***Noida, Uttar Pradesh***

EXPERIENCE

Voxomos — **May 2026 – Present**
AI Engineer Intern — ***Noida, Uttar Pradesh***

- Architected "Jarvis," a **ReAct/LangGraph** agentic AI system that autonomously sends emails, places calls, and checks calendars, with tiered human-approval gating and **Pydantic**-validated guardrails for sensitive actions.
- Engineered an ASR evaluation pipeline benchmarking **Deepgram** and **Azure Speech** against human-annotated Hindi-English transcriptions using WER, CER, BLEU, and BERTScore, with a locally hosted LLM as semantic judge.
- Developed a **FastAPI + MySQL/MongoDB** billing engine computing campaign-wise call statistics via configurable duration slabs; fixed a DB connection leak and added SQL-injection-safe identifier validation.
- Delivered a **React + ECharts** analytics dashboard with interactive charts, rich tooltips, and click-to-detail modals for real-time call campaign reporting.

Entrepreneurship Development Cell (EDC), JSSATEN — **Jan 2024 – Jul 2026**
General Secretary — ***Noida, Uttar Pradesh***

- Owned end-to-end data management and updates for the club website **edcjss.com**, including event pages and registration flows serving **500+** students per cycle.
- Directed data-driven social media campaigns, analyzing weekly engagement metrics and optimizing posting schedules to grow followers by **38%** over **8 months**.
- Tracked and reported event KPIs — attendance rate, registration conversion, and post-event feedback scores — across **10+** annual events.

PROJECTS

Crop Health & Disease Detection | *Python, CNN, FastAPI, Cloudinary* — **Dec 2025 – March 2026**

- Trained a **CNN** classifier on a **50,000+** image dataset, achieving **87% accuracy** across **16** crop disease categories — a **23%** improvement over baseline logistic regression.
- Built a **Global Crop Health Dashboard** with a 3D geolocation globe mapping **1,000+** disease reports by region, reducing time-to-insight for epidemiological trend analysis by **60%**.
- Integrated **Cloudinary** for automated image ingestion; the end-to-end pipeline processes and classifies images in under **2 seconds**, supporting up to **500** concurrent upload requests.

Enterprise AI Workspace | *React, FastAPI, Supabase pgvector, LiteLLM* — **Aug 2026**

- Architected a multilingual **Retrieval-Augmented Generation (RAG)** application to parse, index, and query complex PDF documents using **FastAPI**, **Sentence-Transformers**, and **Supabase pgvector**.
- Integrated **LiteLLM** to route queries across multiple AI providers (OpenAI, Gemini, Groq) and built a SaaS token-tracking system supporting a **5,000-token** IP-based free trial.
- Engineered a **React** frontend featuring browser-native TTS for translated outputs and strict, transparent source citations (raw text hover) to mitigate AI hallucinations.

Folost | *React, PostgreSQL, PostGIS, Express, Cloudinary* — **Jan 2026 – May 2026**

- Created a full-stack **Lost & Found asset recovery platform** with geo-tagged item reporting, using **PostGIS** spatial queries and Leaflet reverse geocoding to improve location-matching precision by **40%** vs. text-only search.
- Structured a **PostgreSQL** schema with **geospatial indexing** supporting **5+** report attributes per record, reducing duplicate report noise by **30%** through a claim-verification flow.
- Implemented a **Cloudinary**-backed image storage pipeline for multi-image uploads per report, cutting average upload time by **45%** and enabling image-based similarity matching for asset recovery.

PUBLICATIONS

IEEE Research Paper: *Plant Disease Detection and Treatment Using Generative AI: A Study on Tomato and Chilli Crops*. Published in the **2026 ICSETI** conference. Co-authored a hybrid CNN/GenAI system achieving **96.8%** real-field accuracy for disease diagnosis and automated treatment recommendations.

TECHNICAL SKILLS

Languages & Frameworks: Python, SQL, JavaScript, Java, React.js, Express.js, Tailwind CSS
Databases & AI/ML: PostgreSQL, MySQL, MongoDB, Supabase, TensorFlow, Keras, Pandas
Developer Tools: Cloudinary, Vercel, Git, GitHub, Postman